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Experiment 1:

AIM: Write a python program to compute Central Tendency Measures: Mean,Median,Mode
Measure of Dispersion: Variance,Standard Deviation.

PROGRAM:

```
import numpy as np
import statistics as st
from math import isnan
from itertools import filterfalse
data=[9,8,5,4,3,7,6,6,7,8,8,8,4,4,2]
print("Dataset: ",data)
mean=st.mean(data)
mode=st.mode(data)
median=st.median(data)
variance=st.variance(data)
std_dev=st.stdev(data)
print("\nCentral Tendency Measures")
print("Mean: ",mean)
print("Mode: ",mode)
print("Median: ",median)
print("Variance: ",variance)
print("Standard Deviation: ",std_dev)
```

Output:

```
Dataset: [9, 8, 5, 4, 3, 7, 6, 6, 7, 8, 8, 8, 4, 4, 2]
Central Tendency Measures
Mean: 5.933333333333334
Mode: 8
Median: 6
Variance: 4.6380952380952385
Standard Deviation: 2.1536237457121516
```

Central Tendency Measures with Missing Data:

PROGRAM:

```
data1=[20.7,float('NaN'),19.2,18.3,float('NaN'),14,4]
sorted(data1)
print(data1)
print(st.median(data1))
```

```
sum(map(isnan,data1))
clean=list(filterfalse(isnan,data1))
print(clean)
print(sorted(clean))
print('median=',st.median(clean))
print('median_low=',st.median_low(clean))
print('median_high=',st.median_high(clean))
print('mode=',st.mode(clean))
print('mode=',st.mode([1,1,2,3,3,3,3,4]))
print('mode=',st.mode(["red","blue","blue","red","green","red","red"]))
print('multimode=',st.multimode('aabbbbbccddddeeffffgg'))
```

Output:

```
[20.7, nan, 19.2, 18.3, nan, 14, 4]
14
[20.7, 19.2, 18.3, 14, 4]
[4, 14, 18.3, 19.2, 20.7]
median= 18.3
median_low= 18.3
median_high= 18.3
mode= 20.7
mode= 3
mode= red
multimode= ['b', 'd']
```

Covariance and Correlation:

PROGRAM:

```
x=[10,15,7,8,8]
y=[40,35,50,50,29]
print('Covariance= ',np.cov(x,y))
print('Corelation= ',np.correlate(x,y))
```

Output:

```
Covariance= [[ 10.3 -12.85] [-12.85 85.7 ]]
Corelation= [1907]
```